

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|-----------------------------|---|
| 1. Scalar _____ | (A) The number of independent columns of a matrix. |
| 2. Magnitude _____ | (B) A vector whose length is exactly 1. |
| 3. Unit vector _____ | (C) All vectors you can reach as $A\mathbf{x}$ — the span of a matrix's columns. |
| 4. Linear combination _____ | (D) A single number that scales (stretches or flips) a vector. |
| 5. Span _____ | (E) $\mathbf{v} \cdot \mathbf{w} = v_1w_1 + v_2w_2$; it measures length and angle. |
| 6. Dot product _____ | (F) The length of a vector, $\sqrt{v_1^2 + v_2^2}$. |
| 7. Column space _____ | (G) A sum of scaled vectors, $c\mathbf{v} + d\mathbf{w}$. |
| 8. Rank _____ | (H) The set of <i>all</i> linear combinations of a set of vectors. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- Two nonzero vectors are **perpendicular** exactly when (A) their sum is $\mathbf{0}$ (B) $\mathbf{v} \cdot \mathbf{w} = 0$ (C) $\mathbf{v} \cdot \mathbf{w} = 1$
(D) they have equal length
- The vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 6 \end{bmatrix}$ span (A) a single point (B) a line (C) the whole plane (D) all of space
- The product $A\mathbf{x}$ is (A) a combination of the *columns* of A (B) a combination of the rows of A
(C) always $\mathbf{0}$ (D) the length of $\mathbf{x}$
- The matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has rank (A) 0 (B) 1 (C) 2 (D) 4
- The unit vector $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ has length (A) 0 (B) 1 (C) $\|\mathbf{v}\|$ (D) $\|\mathbf{v}\|^2$
- A vector $\mathbf{b}$ lies in the column space of A exactly when (A) $\mathbf{b} = \mathbf{0}$ (B) $A\mathbf{x} = \mathbf{b}$ has a solution
(C) $\mathbf{b}$ is a unit vector (D) $\mathbf{b}$ is perpendicular to every column

Part C — Short Answer & Computation (30 pts)

Show your work.

1. Let $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Compute (a) $\mathbf{v} + \mathbf{w}$, (b) $2\mathbf{v} - \mathbf{w}$, (c) $\|\mathbf{v}\|$.
2. Write the unit vector that points the same direction as $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.
3. Let $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{u} \cdot \mathbf{v}$, then state whether $\mathbf{u}$ and $\mathbf{v}$ are perpendicular and how you know.
4. Find weights c and d so that $c\begin{bmatrix} 2 \\ 1 \end{bmatrix} + d\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$.
5. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Compute $A\mathbf{x}$ as a combination of the columns of A .
6. Factor $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}$ as $A = CR$. Which columns are independent? Write C , build the recipe R , and give the rank.

7. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. For each target, say whether it is reachable as $A\mathbf{x}$ (is it in the column space?) and justify in one line. **(a)** $\mathbf{b} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ **(b)** $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. A matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ has columns $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$. **(a)** Is the column space a line or the whole plane? How do you know? **(b)** Is $\mathbf{b} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ reachable? If so, find the weights. **(c)** If the second column were changed to $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$, what would happen to the column space, and why? _____

2. Two viewers' taste profiles are $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{u} \cdot \mathbf{v}$ and explain what the result says about how similar the two viewers' tastes are, using what you know about the sign of a dot product.